The most elegant solutions in machine learning (ML) often emerge from constraint. Working across molecular modeling, computer vision, medical AI, and generative AI, I have consistently faced scenarios where data requirements make traditional supervised learning unrealistic. This has led me to explore the full spectrum of data-efficient learning including transfer learning, bridging sim-to-real gaps, weakly-supervised methods, and reinforcement learning that generates its own training signal. I am interested in developing methods that improve ML performance with limited or low quality labeled data, a challenge critical across academia and industry. This perspective has evolved through my experience at KEF Robotics and my Masters at Tufts University, and now drives my pursuit of a PhD through the **Ellis Program**.

Learning What It Takes to Publish Papers

Molecular science is a domain in which ML can be difficult to apply, primarily due to the cost of gathering high-quality data. At the **University of Pittsburgh** in the Chemical Engineering department, I published several works in this space [1, 2, 3, 4]. This taught me to collaborate across disciplines, iterate on ideas, and explain findings succinctly for publication. In my first-author paper, published in a high-impact journal, I redefined how we quantify molecule-to-molecule interactions in the bond centric model, leading to a 71% reduction in RMSE on our dataset [4]. Through my time at Pitt, I was drawn to the methodological ML work published in ICML, NeurIPS, and ICLR, particularly around learning from limited labeled data, computer vision, and NLP. This passion was why I decided to pivot to computer science and methods oriented work. However, I wanted practical experience before pursuing doctoral research. KEF Robotics offered a chance to rapidly deploy SOTA techniques on challenging problems with tight data constraints.

Transfer Learning, Bridging Sim-to-Real Gaps, and Leadership

At **KEF Robotics**, I developed an approach to multi-modal semantic segmentation for tethered drone navigation under severe data constraints. The critical challenge was detecting power lines (thin, deadly obstacles) across both RGB and infrared modalities with minimal labeled real-world data. I designed a three-stage pipeline: (1) transformer-based segmentation, fine-tuning a source model from COCO using drone simulation data, (2) progressive domain adaptation using limited labeled real-world RGB imagery, and (3) automated cross-modal label transfer using geometric correspondence to generate IR training data from RGB predictions. This approach enabled robust power line detection across both modalities in real-world scenarios without expensive manual IR annotation. I presented this work at XChangeIdeas Pittsburgh [5] where I was selected as one of only two presenters representing KEF Robotics at this invited talk, presenting to an audience of over 100 industry professionals. After this I managed a year long technical project, where I led a team of five engineers, and was the sole machine learning engineer on the project. The project involved on-device object detection with 3D localization, 3D reconstruction, and monocular depth estimation. By the end of the project, we had two successful in person demos and were selected as the drone autonomy group for the next round of funding. The project honed my leadership skills and ability to break down long term projects into accomplished goals. My experience at KEF Robotics confirmed my passion for data-efficient deep learning, but it also helped me realize that I'm more drawn to fundamental research than industry engineering roles.

The Importance of Inductive Bias

To bridge practical implementation skills with rigorous ML theory and academic research experience, I pursued a Master's in Computer Science at **Tufts University**, where coursework in generative models, learning from limited labeled data, computer vision, RL, and algorithms complemented hands-on research. My research with Professor Michael Hughes involves classifying brain scans (MRI and CT) for early indicators of stroke and dementia. This task naturally fits a multiple instance learning (MIL) framework, since diagnostic labels are only available for whole scans (bags) rather than individual slices (instances) due to the high cost of expert annotations. Several MIL methods assumed instances are independent, and more recently methods attempt to learn interactions between instances. Since slices from the same subject's scan are rarely independent and often exhibit shared information that can improve classification accuracy [6], my work in **ML4H 2025** [7] evaluates how well MIL models capture these critical inter-instance correlations. By generating our own synthetic dataset, we constructed a Bayes estimator that serves as an optimal upper bound on model performance. We benchmark

both correlated MIL methods and non-correlated approaches. Our work revealed a generalization gap between correlated and non-correlated MIL architectures, and that narrowing that gap takes much more labeled data than is normally available in the medical domain. This implies that current MIL architectures may not be well suited for brain scan classification and that stronger inductive bias for local interactions is necessary for correlated MIL at realistic dataset sizes.

LLM RL Post-Training, Fine-Tuning with LoRA, Psychology Inspired Experiments

My work with Professor Jivko Sinapov dives into SOTA reinforcement learning (RL) techniques for improving vision-language models' (VLMs) visual reasoning capabilities. I tackled two main applications: First, **SVG Generation from Text**, where open-source LLMs are poor at generating semantically meaningful and aesthetic SVGs from text descriptions. I used group relative policy optimization with a custom 3-part reward encouraging structured outputs, aesthetic images, and semantic alignment to the input text descriptions, to drastically improve Qwen-2.5 7B's SVG generation capabilities (18% improvement on reward) [8]. Second, **Object Counting in vision language models (VLMs)**: I am currently running experiments for a new, exciting project. It is well known that VLMs struggle with the task of counting objects [9]. Psychologists have found that movement, visual markers, and sequential thinking are crucial to humans' ability to count [10]. Can we bake that intuition into a VLM to improve its ability to count as well? Treating counting as not just a look-once-and-return-a-count task, but as a sequential task where the VLM has access to tools for marking up the image for keeping track of objects.

Why the Ellis Program?

[The structure above (project paragraphs anchored in the opening thesis) is the easy part to adapt. The paragraph below is where most of your time should go.]

[Personal hook: 1-2 sentences on *why this program* (for me it was location, exchange model, family ties to Europe)] [Advisor fit: name 2–3 specific advisors, and for each talk about their work, why their lab would be ideal for you, and why you would be ideal for their lab. Find and talk about at least one paper from each lab that you found interesting (explain why).]

The thread connecting my work, from molecular modeling to drone navigation to medical AI and LLM post-training, is the conviction that elegant solutions emerge from constraint. That constraint could be a lack of labeled data, quality data, rich data representations or inherent model limitations. My extensive experience with computer science and machine learning related research in both academia and industry, along with my leadership experience and strong desire to learn and grow, would make me well prepared for a PhD through the **Ellis Program**. After my PhD, I would like to continue a research career, either in academia or at a research lab in industry.

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